

Clinton Power Station
8401 Power Road
Clinton, IL 61727



Exelon Generation®

U-604422
July 16, 2018

10CFR50.73
SRRS 5A.108

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2017-002-02

Enclosed is Licensee Event Report (LER) 2017-002-02: Failure of the Division 1 Diesel Generator Ventilation Fan Load Sequence Relay Circuit During Concurrent Maintenance of RHR Division 2 Results in an Unanalyzed Condition. This is a supplemental report to LER 2017-002-01 submitted to the NRC on July 5, 2017. The updated information in the LER is denoted by revision bars located in the right-hand margin. This report is being submitted in accordance with the requirements of 10 CFR 50.73. There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. Dale Shelton, Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Theodore R. Stoner
Site Vice President
Clinton Power Station

GS/lam

Attachment: Licensee Event Report 2017-002-02

cc:

Regional Administrator— NRC Region III
NRC Senior Resident Inspector - Clinton Power Station
Office of Nuclear Facility Safety — Illinois Emergency Management Agency

TEZZ
NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollect.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Clinton Power Station	2. Docket Number 05000461	3. Page 1 OF 4
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4. Title

Failure of the Division 1 Diesel Generator Ventilation Fan Load Sequence Relay Circuit During Concurrent Maintenance of RHR Division 2 Results in an Unanalyzed Condition

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
03	09	17	2017	- 002	- 02	07	16	18	Facility Name	Docket Number
										050000
										050000

9. Operating Mode

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☒ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
099	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact

Dale A. Shelton, Regulatory Assurance Manager

Telephone Number (Include Area Code)

217-937-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
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14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On Tuesday, March 7, 2017 at 2258 CDT, an Equipment Operator detected a relay cycling every 10 seconds from the Division (Div.) 1 480V Unit Substation (Sub) 1A. Agastat Time Delay Relay (TDR) 427X2-41A (X2 relay) was determined to be the source of the clicking sound. The X2 relay supports the undervoltage load shed and restoration of the Div. 1 Emergency Diesel Generator (EDG) ventilation room fan 1VD01CA. The logic is designed to shed 1VD01CA on undervoltage, prevent it from restarting, then allow it to be restored 10 seconds after voltage is restored either by the EDG or return of the safety bus. An investigation of the relay cycling condition determined that 1VD01CA was unable to respond to a demand signal and the cycling was the effect of interaction between the X2 and 427X3-41A (X3) relays. As a result, Div. 1 EDG was declared inoperable on March 9. With the Div. 2 Residual Heat Removal (RHR) System already inoperable due to scheduled maintenance, the plant was determined to be in an unanalyzed condition. A causal analysis determined that the original design of the EDG ventilation fan relay logic relied on executing parallel sequencing logic to perform the control function, which was vulnerable to differences in relay model drop out voltages. Corrective actions included the replacement of the X3 relay with the original design relay and subsequent circuit modifications to eliminate the vulnerability to lower drop out voltage settings. Div. 2 RHR was returned to service on March 8. Subsequently, Div. 1 EDG was restored to OPERABLE on March 11. This is reportable as an unanalyzed condition, operation prohibited by the Technical Specifications and a loss of safety function.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Clinton Power Station	05000461	2017	- 002	- 02

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric—Boiling Water Reactor, 3473 Megawatts Thermal Rated Core Power
Energy Industry Identification System (EIS) codes are identified in the text as [XX]

EVENT IDENTIFICATION

Failure of the Division 1 Diesel Generator Ventilation Fan Load Sequence Relay Circuit During
Concurrent Maintenance of RHR Division 2 Results in an Unanalyzed Condition

A. Plant Operating Conditions before the Event

Unit: 1	Event Date: March 9, 2017	Event Time: 0319
Mode: 1	Mode Name: Power Operation	Reactor Power: 99 percent

B. DESCRIPTION OF EVENT

On March 7 at 2258, an area operator detected a "clicking sound" coming from safety related Unit Substation 1A (1AP11E). The sound was determined to be emanating from relay 427X2-41A (X2 relay) which was cycling every ten (10) seconds. The model of this component is an Agastat Time Delay Relay (TDR) that provides the signal to reset the Load Shed and Resequencing circuit for Div. 1 Emergency Diesel Generator (EDG) Room Vent fan 1VD01CA. Div. 1 EDG was declared inoperable on March 9 when it was determined that 1VD01CA was unable to respond to a demand signal.

An event investigation determined that relay 427X3-41A (X3 relay) in the undervoltage circuitry for 1AP11E was not operating as expected, causing the cycling of the X2 relay. The effect of this condition was that (1) the associated Fan 1VD01CA would be unable to start automatically or manually when required, and (2) the ability of the Div. 1 fan to properly sequence following restoration from a bus undervoltage condition was impacted. During an assessment of relay replacement history, it was established that the X3 relay which was originally an ITE Gould J13 relay had been replaced with a General Electric CR120B relay in 2008. The replacement X3 relay combined with the installed X2 relay satisfactorily passed post maintenance testing in 2008 (including undervoltage functional testing) and also passed integrated surveillance testing in 2009. A "like-for-like" replacement of the X2 relay was performed in 2011 and passed integrated surveillance testing in 2011 and 2013. The investigation determined, however, that the combined operating characteristics of the X3 and X2 relays was adversely impacted by a second "like-for-like" replacement of the X2 relay in May 2016. Through an error in work order planning, no testing was specified for this "like-for-like" replacement. However, the impact on the operating characteristics of the electrical circuit for this X2 replacement would not have been detected by the procedurally required post maintenance testing. Due to the successful integrated testing since 2008, variations in the dropout voltage and associated impacts of the relays used in this circuit had not been previously recognized.

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NARRATIVE

The combination of replacement of the X3 (ITE Gould J13) relays with CR120B relays in 2008 and the perceived "like-for-like" replacement of the Agastat X2 relay in May 2016 established a relay coordination condition where the required relay sequence of operation was adversely affected due to variations in the manufacturer's electrical characteristics of the relays supplied for the electrical circuit.

Following the replacement of the CR120B relay with the original ITE Gould J13 relay design and successful undervoltage testing, the Div. 1 EDG was restored to OPERABLE status.

C. CAUSE OF EVENT

The cause of the event was that the original design of the EDG ventilation fan relay logic relied on executing parallel sequencing logic to perform the control function, which was vulnerable to differences in relay model drop out voltages. As the critical characteristics of the fan logic were not clearly documented or known, it was assumed by the procurement engineer that differences in relay drop out voltages were not material to the design function.

D. SAFETY ANALYSIS

There were no actual safety consequences associated with the event described in this report since offsite power remained available during the event and no other events occurred necessitating ECCS actuation. Plant systems necessary to conduct a safe and orderly shutdown remained available during this event. The Division 2 EDG, although inoperable intermittently to perform required surveillance testing would have been able to be restored expeditiously in the event of an accident in conjunction with a loss of offsite power.

The event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition which was prohibited by the plant's Technical Specifications. This event is also reportable in accordance with 10 CFR 50.73(a)(2)(ii)(B) as a condition that resulted in the plant being in an unanalyzed condition that significantly degraded plant safety. Additionally, a review of plant logs identified occurrences when the Div. 2 EDG was made inoperable to support surveillance testing during this time period. As a result, this event is also reportable as a loss of safety function per 10 CFR 50.73 (a)(2)(v)(D) which is considered a safety system functional failure.

E. CORRECTIVE ACTIONS

The corrective actions taken include replacing the existing CR120B relay with an original ITE Gould J13 relay design and successfully performing the undervoltage test. A circuit modification has been completed to eliminate the vulnerability to the lower dropout voltage settings expected from use of the CR120B relays. A procurement engineering standard has been developed to address complex relay logic schemes and what critical characteristics should be selected for evaluation when performing a safety basis, a commercial grade dedication, and/or item equivalency evaluation for complex circuits involving relay coordination.



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NARRATIVE

F. PREVIOUS SIMILAR OCCURENCES

No previous occurrences involving similar circuits and relay interactions have been identified.

G. COMPONENT FAILURE DATA

No CPS component failures associated with this event were identified.